Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 1 of 12 · Question 5 part c.i · 4 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

5 (a) (i)		• alternative form of gene; • only expressed in homozygote/when 2 recessive alleles are present/ when no dominant allele is present/not expressed in heterozygote/;		2
(b)		any two from • person 2; • person 4; • person 9;		
	(ii)	Aa;		1
(c)	(i)	parents Aa × Aa; gametes A a A a; F1 AA 2Aa aa; phenotype normal normal carrier sickle-cell;	2 marks max for unlabelled Punnett square (mp's 2 and 3)	4
	(ii)	• chance of a boy = 0.5/50%; • chance of sickle cell anaemia = 0.25/25%; • chance of boy with condition = 0.5 × 0.25 OR $\frac{50 \times 25}{100}$; • = 0.125/12.5%;		4

Total 13 marks

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 1 of 12 · Question 5 part c.i · 4 marks

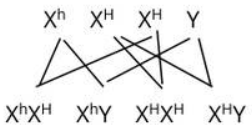
2023 NOV · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 2 of 12 · Question 9 part c.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part c.i in this mark scheme extract

9	(a)	(i)	A (the inheritance of two different alleles, both of which are expressed) B is incorrect as one allele is not expressed in codominant inheritance C is incorrect as the ABO inheritance does not involve the inheritance of multiple alleles D is incorrect as ABO inheritances does not involve the inheritance of multiple alleles		1
		(ii)	I^AI^A and I^AI^o		1
	(b)	(i)	• blood group O contains no antigens; • antigens removed/digested (by enzyme); • carbohydrase digest carbohydrates; • protease digest proteins;		Max 3
		(ii)	reduces need to find donor/cheaper/blood readily available/less chance of transfusing incorrect blood group/less risk of disease/infection;		1
	(c)	(i)	parent genotypes X^hX^H and X^HY; gametes and linkage X^h X^H X^H Y  ;	One mark for parent genotypes and one mark for gametes **and** linkage	1 1
		(ii)	1 in 4/25%/quarter/1/4 /1:3/0.25;		1

Total question 9 = 9 mark

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 2 of 12 · Question 9 part c.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part c.i in this mark scheme extract

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Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 3 of 12 · Question 8 part b.i · 4 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

8	(a)	any three from • change in base sequence/named mutation (1) • of DNA (1) • causing change in genotype/gene (1) • incorrect protein produced (1) • change in phenotype/characteristics (1)	Allow change in DNA for mps 1 and 2 (1)	3
	(b) (i)	father X^HY mother X^HX^h (1) gametes X^H Y X^H X^h (1) F1 X^HX^H normal female; X^HX^h carrier female; X^HY normal male; X^hY haemophiliac male (1)		4
	(ii)	(chance of first and second child is) ¼/0.25/25% (1) for both is ¼ x ¼/0.25 x 0.25 (1) = 1/16/0.0625/6.25% (1)		3
	(c)	any five from • males only have one X chromosome (1) • two X chromosomes/H and h needed to be a carrier (1) • recessive allele/h (1) • carried on X chromosome (1) • either present or not (1) • no chance for dominant allele to be present (1) • to mask recessive (1)		5

Total 15 marks

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 3 of 12 · Question 8 part b.i · 4 marks

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Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 4 of 12 · Question 8 part b.i · 4 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
8 (a)	(i)	absorbs uv light; reduces/prevents risk of skin cancer;	Allow protects against (UV) radiation	2
	(ii)	eye/ retina/ choroid/ red blood cell/ haemoglobin ;		1
(b)	(i)	mother x father Hh hh; gametes H h h; fertilisation Hh hh; phenotype white forelock normal;	Reject diagrams that show sex linkage	4
	(ii)	chance of producing a boy is 0.5/½; chance of producing offspring with condition 0.5/½; chance of producing boy with condition is ½ × ½ = ¼;	Allow 50% or equivalent anywhere in calculation Full marks for correct final answer	3
(c)		Any 4 from different (genetic) code produced; different order of amino acids; codes for different proteins/enzymes; enzyme substrate complex not formed; causes change in pigment;	Allow change in base sequence/genetic code Allow any reference to change in amino acids	4

Total 14 marks

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 4 of 12 · Question 8 part b.i · 4 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

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Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 5 of 12 · Question 7 part b.ii · 6 marks

2022 JAN · Paper 02 · find part b.ii in this mark scheme extract

number				
7	(a) (i)	• appears in both males and females; • in similar numbers;		2
	(ii)	• condition present in heterozygous genotype/only one allele needed/affected are carriers; • only some children (from affected and unaffected parents) are affected/not affected;		2
	(b) (i)	• probability of a boy is 0.5/50%; • probability of no Huntington's is 1.0; • overall probability is 0.5 × 1.0 = 0.5/50%/1;		3
	(ii)	person 3 must be hh/homozygous recessive; does not carry a faulty allele; person 2 is Hh/heterozygous/carries a faulty allele / carries one normal allele; as father not affected; offspring either Hh or hh; half with condition and half without/50% chance of being affected;	marks can be awarded for suitable genetic crosses that are annotated	6

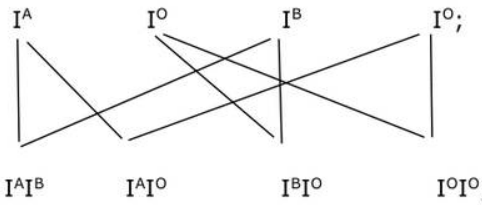
Total 13 marks

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Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 1 of 7 · find part a.iii in this mark scheme extract

4 (a)	(i)	**Africa** Blood group / Percentage of blood group O: 68 A: 17/17.5 B: 10; AB: 5 **Europe** Blood group / Percentage of blood group O: 45 A: 40 B: 12 AB: 3;		2
	(ii)	• reference to antigens on red blood cells (group B); • antibodies produced; • causing agglutination/clumping (of red blood cells);		3
	(iii)	I^AI^O I^BI^O;  I^AI^B I^AI^O I^BI^O I^OI^O;	Full marks for a Punnett square if parent genotypes are given.	3
	(iv)	1 in 4:/25%/¼/0.25;		1
(b)		• <u>deoxygenated</u> blood moves into the left side/ventricle of the heart; • (deoxygenated blood) mixes with <u>oxygenated</u> blood; • less oxygen travelling around the body/to body/cells/in bloodstream; • breathes faster to get more oxygen into the body;		4
				Total 13

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 2 of 7 · find part a.iii in this mark scheme extract

	Answer	Notes	Marks

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 3 of 7 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks
6	Any six from the following: • remove plasmid from bacteria; • locate the human insulin gene • cut/remove insulin gene (from human); • cut open the plasmid; • using restriction enzymes; • add the human insulin gene to the plasmid; • use of ligase; • ref recombinant DNA; • return the plasmid to a bacterial cell;	Suitably drawn **and** annotated diagrams can score full marks.	6
			Total 6

Question number	Answer	Notes	Marks
7 (a) (i)	obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH;		1
(ii)	sucrose;		1
(iii)	any three from • add sodium alginate with distilled water; • mix (with a hand blender); • until all sodium alginate has dissolved; • add calcium chloride; • wash beads;		3
(b)	**B** insulin is released from the pancreas and travels to the liver **A** is incorrect as glucagon does not convert glucose to glycogen **C** is incorrect as glucagon is not released from the liver **D** is incorrect as insulin is not released from the liver		1]

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 4 of 7 · find part a.iii in this mark scheme extract

(c) (i)	• (66.65 ÷ 100) = 0.6665; • (x6) = 4.0 (million);	Allow 0.67 Allow 3.999	2
(ii)	• 38 49 51 62 65 78 82 91 96; • 65;		2
			Total 10

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 5 of 7 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks
8 (a)	**Advantages embryonic stem cells:** • generate most cell types; • can maintain and grow for long lengths of time in culture; • useful for drug development studies/Parkinson's disease; **Disadvantages of embryonic stem cells:** • can cause tumours/cancer; • many embryos destroyed; • some people think it is unethical/killing/murder; • possibility of rejection if used in transplants **Advantages of adult stem cells:** • can be reprogrammed; • less likely rejected if used in transplants; • taken from the patient itself; **Disadvantages of adult stem cells:** • limited ability to differentiate/develop into different types of cell; • cannot be grown for long periods of time in culture; • usually a very small number found in body tissues; • can only grow small numbers;	Three marks from embryonic stem cell section and three marks from adult stem cell section Allow a valid named ethic	6
			Total 6

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 6 of 7 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks
9 (a)	• beta blockers; • block/bind to receptors; • which prevents the release of adrenaline/noradrenaline; • heart rate decreases; OR • ACE inhibitors; • reduce the activity of enzymes; • less angiotensin; • less narrowing of blood vessels;		4
(b)(i)	any two from • how much pressure blood is exerting; • against artery walls; • when ventricles contract/pump;		2
(ii)	• pressure in the arteries; • when the heart/ventricles at rest/relaxed/between beats;		2
			Total 8

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 6 of 12 · Question 4 part a.iii · 3 marks

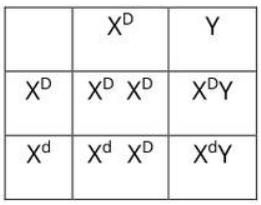
2022 JUN · Paper 01 · extract 7 of 7 · find part a.iii in this mark scheme extract

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Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 7 of 12 · Question 8 part b.iii · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

8 (a)	any 4 from • change in base sequence produces different mRNA; • altered codon (or codons); • so different amino acids coded; • incorrect protein made; • could produce a stop codon so no protein made/faulty protein made;		4
(b) (i)	• faulty allele present on one chromosome only; • effects overridden by a dominant allele;		2
(ii)	• one male in generation 1 has DMD; • (faulty) allele must be on X chromosome; • male is XY/does not carry dominant allele on Y chromosome; • carriers present;		3
(iii)	• correct parental genotypes; • correct genotypes for generation 1; 		2
			Total 11

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 7 of 12 · Question 8 part b.iii · 2 marks

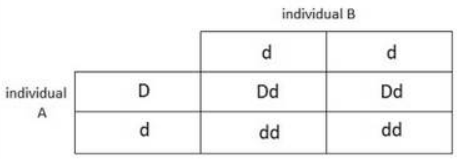
2022 JUN · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 8 of 12 · Question 7 part c.ii · 4 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

Number				
7	(a)	Any three from: • changes the DNA/base/codon sequence; • changes the order of the amino acids/different amino acid coded for; • changes the function/shape of the protein/different protein; • affects a characteristic or function/genotype in the individual;		3
	(b) (i)	Two from: • dialysis has to be completed several times a week/transplant permanent; • increased life expectancy; • less restrictions on diet; • better quality of life/freedom to move around/normal lifestyle;		2
	(ii)	100 x 3233÷4733; = 68.3/68		2
	(iii)	Two from: • people have two kidneys; • two recipients from one donor; • living donation/can live with just one;		2
	(c) (i)	0.5:1;	Accept 1:2	1
	(ii)	 • correct genotype individual A; • correct genotype individual B; • 50% offspring heterozygous/correct genotype and affected; • 50% homozygous recessive/correct genotype and not affected;		4

Total 14 marks

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 8 of 12 · Question 7 part c.ii · 4 marks

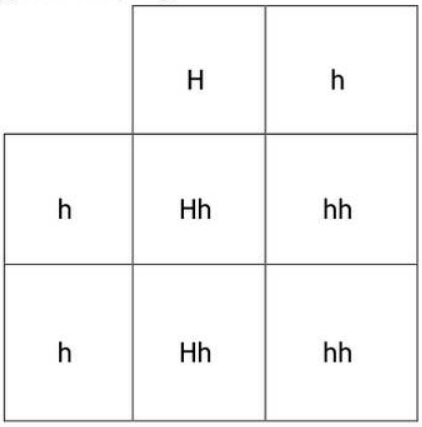
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Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 9 of 12 · Question 3 part b · 2 marks

2024 JUN · Paper 02 (regional R) · find part b in this mark scheme extract

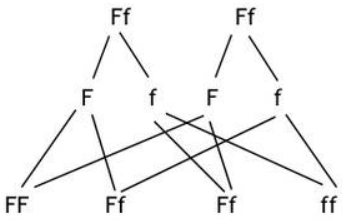
3 (a)	change in the order of bases in DNA;	Allow a description of a type of mutation	(1)
(b)	parent gametes; genotypes of offspring;  	Allow genetic cross Also allow different orientation of alleles in Punnett Square	(2)
(c)	**Advantages** • no (costly) visits to hospital; • no inconvenience of dialysis; • improved quality of life/ longer life/long term solution; **Disadvantages** • reference to difficulty in finding a donor; • rejection/reference to anti-rejection medication for life; • reference to risks associated with surgery;	any one from advantages and any one from disadvantages	(2)

Total for Question 3 = 5 marks

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 10 of 12 · Question 8 part a.iv · 4 marks

2024 JUN · Paper 02 · find part a.iv in this mark scheme extract

Number				
8	(a) (i)	**B** individual 3 The following are incorrect because: **A** individual 1 is heterozygous **C** Individual 5 is heterozygous **D** individual 3 is homozygous dominant		(1)
	(ii)	**D** 4 The following are incorrect because: **A** there are more than 1 heterozygous genotypes **B** there are more than 2 heterozygous genotypes **C** there are more than 3 heterozygous genotypes		(1)
	(iii)	• they have a dominant allele; • over-rides/ masks the effect of the recessive allele/ must have two recessive alleles for the condition to show/ must have homozygous recessive alleles ;	do not allow dominant allele overpowering the recessive allele	(2)
	(iv)	• heterozygous parent genotypes; • gametes; • genotypes of offspring; • offspring genotypes linked to the phenotype for the disease and showing that showing that 1 of the 4 genotype has the disease;  	mps 2,3 and 4 marks for correct Punnett square showing all details	(4)
	(b)	• reference to use of restriction enzyme • (to obtain) the normal/ non faulty/ dominant allele); • use of virus/ correctly named virus; • inserts allele into person **cells**; • functional/normal protein produced;	Accept liposomes ignore plasmids Penalise one for using genes instead of alleles.	(3)

Total for Question 8 = 11 marks

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 11 of 12 · Question 6 part b · 4 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

6 (a) (i)	2500 males; $\frac{2500 \times 0.8}{100}$; = 20;		ecf if use 5000 as starting figure 40 scores 2 marks any number×0.8=1	3
(ii)	small number/ not enough tested;			1
(iii)	• many more males than females with condition; • show that it is sex-linked; • to X chromosome; • females are XX /males one X chromosome/XY • females can be carriers(so fewer)/males cannot be carriers(so more); • (relatively) small number (in whole population)/small chance of being colour blind; • indicates that it is recessive;			5
(b)	parents	$X^BY \times X^BX^b$;		4
	gametes	X^B Y X^B X^b;		
	offspring genotypes	$X^B X^B$ X^BX^b X^BY X^bY;		
	offspring phenotypes	normal carrier/ disease female normal female male red-green male;	If parents are incorrect ecf for mp 2,3 and 4 If X and Y are not used- max of 1 mark	
				Total = 13

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 11 of 12 · Question 6 part b · 4 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 12 of 12 · Question 2 part b.i · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
2	(a)	(i)	movement/swimming;		1
		(ii)	C; A, B and D do not release energy		1
		(iii)	three from: • produces a haploid cell / reduces the number of chromosomes by half/to 23; • so at fertilisation the diploid/full number of/46 chromosomes is restored; • prevents doubling of chromosomes at fertilization; • introduces (genetic) variation;		3
	(b)	(i)	sperm egg X / Y X / XX / XY X / XX / XY ;	One for egg One for sperm One for the offspring	3
		(ii)	Two from: • acrosome; • contains enzymes; • digests/breaks down the egg cell membrane;		2

Total 10 marks

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 12 of 12 · Question 2 part b.i · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks